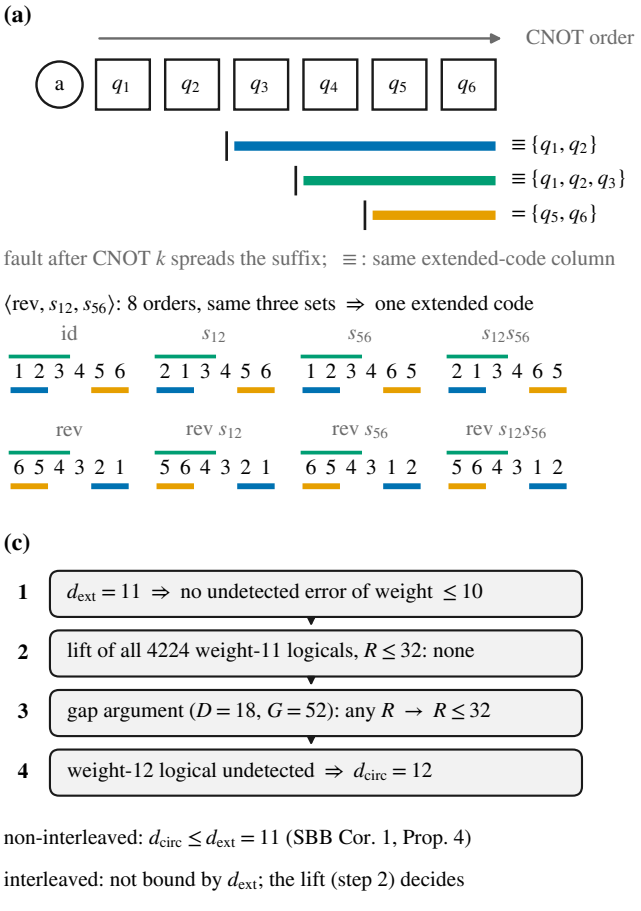
(a)
CNOT order
a
q_1
q_2
q_3
q_4
q_5
q_6
$\equiv \{q_1, q_2\}$
$\equiv \{q_1, q_2, q_3\}$
$= \{q_5, q_6\}$
fault after CNOT k spreads the suffix; $\equiv$: same extended-code column
$\langle \text{rev}, s_{12}, s_{56} \rangle$: 8 orders, same three sets $\Rightarrow$ one extended code
id
1 2 3 4 5 6
s_{12}
2 1 3 4 5 6
s_{56}
1 2 3 4 6 5
$s_{12}s_{56}$
2 1 3 4 6 5
rev
6 5 4 3 2 1
rev s_{12}
5 6 4 3 2 1
rev s_{56}
6 5 4 3 1 2
rev $s_{12}s_{56}$
5 6 4 3 1 2
(c)
1 $d_{\text{ext}} = 11 \Rightarrow$ no undetected error of weight ≤ 10
2 lift of all 4224 weight-11 logicals, $R \leq 32$: none
3 gap argument ($D = 18$, $G = 52$): any $R \rightarrow R \leq 32$
4 weight-12 logical undetected $\Rightarrow d_{\text{circ}} = 12$
non-interleaved: $d_{\text{circ}} \leq d_{\text{ext}} = 11$ (SBB Cor. 1, Prop. 4)
interleaved: not bound by d_{ext}; the lift (step 2) decides
(b)
Fig. 12: undetected
period 18: detected
hook
data 12
data 6
data 0
hook
late
Z_{15}
Z_{18}
Z_{12}
Z_6
early
0 4 8 12
tick ($R = 1$)
0 6 12 18
tick ($R = 1$)